Human Activity Classification Based on Radar Signatures

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I. Introduction

Classifying human activity from their radar signatures has various use cases in health, security, and management. In this report, methods and results that utilize a Support Vector Machines (SVMs) classifier with various hand-crafted features are described. The work in [1] was used as a reference and guide for the methods used in this study.

II. Methods

The opted method uses hand crafted features over a data-driven approach due to the limited amount of data available in combination with the large intra-class variety of the spectrograms. The high data requirements for a data-driven model could thus not be met. Another important consideration was to have good interpretability. Using hand crafted features allows for a better understanding of any successes and failures of the model compared to a data driven approach.

A. Calculation of spectrogram

The original data is from a modulated frequency radar. The range-time diagram is first obtained through a Fast Fourier Transform (FFT), and a moving target indication filter is applied to remove static clutter. The spectrogram was calculated by summing the spectrograms obtained from the selected range bins where activity was detected. A window length of 200 samples, an overlap of 95%, and zero-padding equal to four times the window length were used.

B. Calculation of features

The calculation of features was done for each file independently. The envelope of the signal in each spectrogram was calculated by first cleaning the spectrogram of the noise and then identifying the 5th and 95th percentile of the total signal for each time bin. Activity in each time bin was determined using a combination of the envelope bandwidth, the maximum detected velocity, and the total signal energy within that bin.

This yielded features such as the maximum velocity indicated by the envelope, the mean velocity indicated by it, the max and mean bandwidth of the signal, how long the activity lasted and what part of the spectrogram this covered and how long the longest active section of the activity lasted.

Secondly, the signal was summed along the time axis to obtain a distribution of signal strength over the Doppler shifts. The central region containing most of the signal energy was selected and normalized by the total signal within that region. From this normalized Doppler distribution, several statistical descriptors were extracted, including variance, skewness, kurtosis, and the weighted standard deviation and mean distance from zero Doppler.

Additional features included the mean of the total signal per time divided by the maximum signal of that time bin, the normalized difference between the positive and negative signal, the standard deviation of the range at which the signal was maximal, obtained from the range-time diagram, the mean cadence obtained from the cadence velocity diagram and the average velocity weighted by signal.

C. Feature selection

Many of these features were naturally highly correlated. Therefore, a sequential forward feature selection algorithm was used to identify which additional feature improved the classification performance the most. To determine the appropriate number of features, the model performance was evaluated while gradually increasing the feature set size. The results indicated that a set of five features produced the best overall performance, with no further improvement observed when additional features were included. Since adding more features can also increase the risk of overfitting and reduce the model's ability to generalize, only five features were retained.

The selected features were 1) Maximum Velocity Bandwidth, 2) Active Motion Fraction, 3) Energy Weighted Velocity, 4) Skewness Doppler Distribution, 5) Total Signal Over Max. 1) describes the largest velocity spread observed in the spectrogram at any time. It captures how wide the motion signature becomes during the most dynamic part of the activity. 2) describes the fraction of time bins in which meaningful motion is detected. It indicates how much of the recording contains active movement. 3) describes the average signed velocity of the spectrogram, weighted by signal energy. It captures whether the dominant motion is mainly in the positive or negative Doppler direction. 4) describes the asymmetry of the normalized Doppler signal distribution. It indicates whether the signal is evenly distributed or concentrated more unevenly across the Doppler range. 5) describes how spread out the Doppler signal is compared with the strongest Doppler component in each time bin. Higher values indicate that the signal is distributed over a wider range of Doppler velocities.

D. Training

The utilized model is a SVM, utilizing an RBF kernel function. For the multiclass problem the algorithm utilizes a one-to-one testing approach, deciding for an observation to which class it likely belongs between each combination of two classes and selecting the class most common.

The utilized data split was a roughly 70/30 training-test split, with the testing being from different sessions. The split is made in such a way that the model classifies the activities of subjects it hasn't seen before based on the activities of the subjects it was trained on. This means that the model learns the patterns of a movement rather than how a specific person performs given movements. The parameters for the SVM were tuned using cross-validation and a hyperparameter grid search.

III. Results

Evaluating the performance of the model on the unseen test data resulted in an overall accuracy of 92.3%. Although the model reached a high overall accuracy, the precisions for classes 4 (Bending and picking up an object) and 5 (Drinking from a cup) were notably lower with 86% and 82% respectfully. The model struggled the most to correctly classify those activities as they present highly variable characteristics in their spectrograms and depend heavily on how the subject realizes the movement.

For classes 1, 2, 3, and 6, the respective precisions were 100%, 96%, 92%, and 98%, indicating that the model performs well on those classes. The best results were obtained for the walking back and forth and falling down classes which were also the classes with the most distinct spectrograms.

IV. Discussion & Conclusions

The results show that the selected radar features provide a strong basis for human activity classification. The high performance on the unseen test set indicates that the model generalizes well beyond the training data and that the extracted features capture relevant differences between the activity classes. In particular, walking back and forth and falling down were classified with very high precision, which is consistent with their more distinctive radar signatures. Walking produces repeated and sustained motion, while falling is characterized by a short but strong velocity response.

The lower precision for bending and drinking suggests that these activities are more difficult to distinguish using the current feature set. Both activities can vary considerably between subjects and may produce less consistent spectrogram patterns. This indicates that the model performs best when activities differ clearly in motion intensity, duration, or velocity distribution, while more subtle activities remain challenging.

Overall, the classification results are promising, with an unseen test accuracy of 92.3%. The selected feature set and tuned SVM classifier are effective for separating most activities, especially those with clearly distinct motion characteristics. Future works could focus on adding more advanced features specifically designed to distinguish between such similar classes and more generalization with motions observed from different angles. Another approach can also extend this study using a data driven approach which could potentially obtain better classification results.

References

[1] Y. Kim and H. Ling, "Human activity classification based on micro-doppler signatures using a support vector machine," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 47, no. 5, pp. 1328–1337, May 2009.